

DriftTrace IDS2018 — Results Summary

All 19 experiment versions vs paper claims

Nadeem Yousaf — NIMBUS Research Centre, MTU Cork, Ireland
August 2026

Paper Claims (IDS2018)

Method	Precision	Recall	F1	Insp. Effort
DriftTrace (paper claim)	0.98 ± 0.01	0.97 ± 0.03	0.97 ± 0.02	0.98 ± 0.02
Vanilla AE (paper baseline)	0.69 ± 0.08	0.96 ± 0.04	0.84 ± 0.04	1.34 ± 0.11
TRANSCENDENT (paper baseline)	0.61 ± 0.11	0.89 ± 0.17	0.75 ± 0.04	1.65 ± 0.26

Explanation (IDS2018-Infiltration, Table VIII + Table VI):

$d_{\text{final}} = 0.0325 \pm 0.1024$ | Boundary crossing = 83.65%

Detection Results — All Versions

File	V	Avg F1	Benign F1	Bot F1	DoS-Hulk F1	Infil F1	Δ Paper
phase2_v1 + phase3_v1/v2	V 1	0.603	0.655	0.554	0.996	0.206	−0.367
phase2_v3 + phase3_v3	V 3	0.573	0.653	0.537	0.983	0.118	−0.397
phase2_v4 + phase3_v4	V 4	~0.45	—	—	—	—	−0.520
phase2_v5 + phase3_v5	V 5	0.531	0.647	0.555	0.718	0.205	−0.439
phase2_v6 + phase3_v6	V 6	0.531	0.647	0.555	0.718	0.205	−0.439
phase2_v7 + phase3_v7	V 7	0.568	0.745	0.603	0.718	0.205	−0.402
phase2_v7 + phase3_v8	V 8	0.478	0.738	0.584	0.456	0.134	−0.492
phase2_v10 + phase3_v10	V 10	0.842	0.891	0.878	0.960	0.641	−0.128
phase2_v10 + phase34_v11	V 11	0.849	0.891	0.872	0.994	0.638	−0.121

File	V	Avg F1	Benign F1	Bot F1	DoS-Hulk F1	Infil F1	Δ Paper
phase2_v10 + phase34_v12	V12	0.851	0.933	0.846	1.000	0.624	-0.119
Paper (DriftTrace)	—	0.970	~0.97	~0.97	~0.97	~0.97	—

★ Gold = recommended versions (V10/V11/V12) | Green = paper target

Explanation Results — Infiltration Hold-out

Paper target: d_final=0.0325, boundary crossing=83.65%

File	V	d_final	Std	Dist. reduction	Boundary crossing
phase3_v10.py	V10	0.0200	—	~85%	40.5%
phase34_v11.py	V11	0.0097	0.0074	66.4%	14.0%
phase34_v12.py	V12	0.0146	—	75.6%	3.0%
Paper	—	0.0325	0.1024	—	83.65%

Explanation Results — All Hold-outs (V11)

Hold-out	d_original	d_final	Reduction	Crossing	Paper target
Benign	~1.20	0.0025	95.9%	98.0%	—
Bot	~0.35	0.0290	92.6%	0.0%	—
DoS-Hulk	~0.40	0.0351	88.4%	0.0%	—
Infiltration	~0.43	0.0097	66.4%	14.0%	0.0325 / 83.65%

Key Diagnostic Measurements

Measurement	Value	Implication
Benign–Infiltration centroid dist.	0.7272	35% closer — hardest to separate
Benign–Bot centroid dist.	1.1205	Normal — converges well
Benign–DoS-Hulk centroid dist.	1.1602	Normal — converges well
Dead features (zero variance)	8 of 83	No gradient signal from these cols
Effective features	75 of 83	Dead cols zeroed by MinMaxScaler
Required magnification (m=10)	13.7×	Infeasible in 300 epochs
Required magnification (m=5)	6.85×	Achievable — all 4 converge